```
import pandas as pd
import matplotlib.pyplot as plt
```

```
from google.colab import files
uploaded = files.upload()
```

Choose files Apex_Retail..._Clean.xlsx

Apex_Retail_Business_US_Clean.xlsx(application/vnd.openxmlformats-officedocument.spreadsheetml.sheet) - 14610 bytes, last modified: 21/10/2026 - 100% done

```
Saving Apex_Retail_Business_US_Clean.xlsx to Apex_Retail_Business_US_Clean.xlsx
```

```
df = pd.read_excel("Apex_Retail_Business_US_Clean.xlsx")
```

```
df.head()
```

	Order_ID	Order_Date	Customer_ID	Customer_Name	City	Region	Category	Sub_Category	Quantity	Unit_P
0	ORD0001	2025-01-01	C144	Customer_22	Chicago	Midwest	Electronics	Binders	7	
1	ORD0002	2025-01-02	C375	Customer_7	New York	East	Electronics	Binders	3	
2	ORD0003	2025-01-03	C220	Customer_24	Los Angeles	West	Furniture	Phones	1	
3	ORD0004	2025-01-04	C838	Customer_29	Houston	South	Furniture	Chairs	5	
4	ORD0005	2025-01-05	C779	Customer_13	New York	East	Electronics	Tables	2	

Next steps: Generate code with df | New interactive sheet

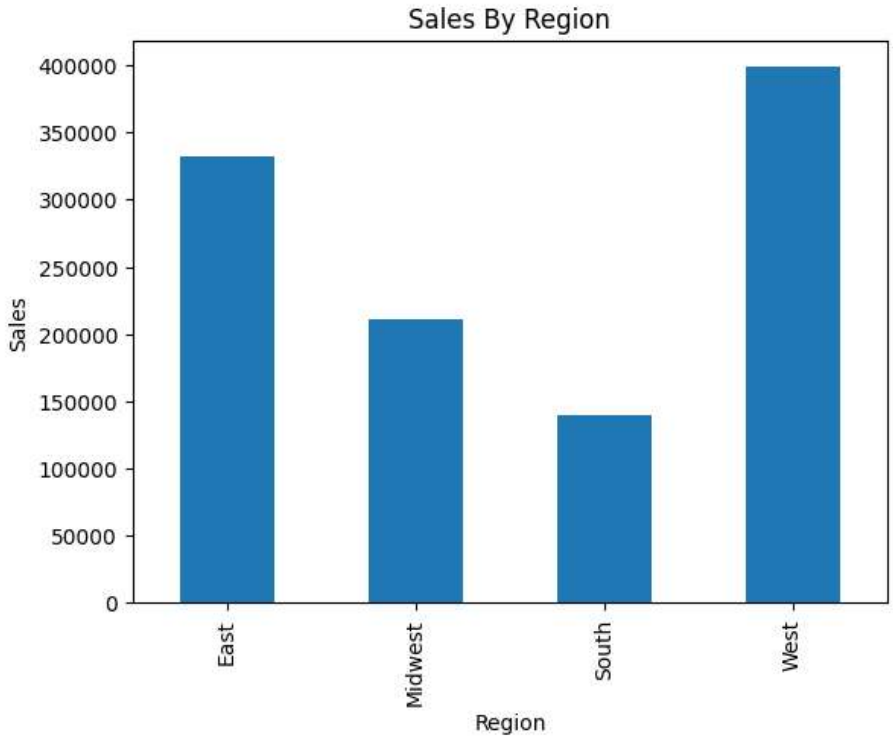
Q.1 Which region generated the highest sales?

```
region_sales = df.groupby("Region")["Sales"].sum()
region_sales.plot(kind="bar")
plt.title("Sales By Region")
plt.xlabel("Region")
plt.ylabel("Sales")
plt.show
```

```
matplotlib.pyplot.show
def show(*args, **kwargs) -> None
```

```
/usr/local/lib/python3.13/dist-packages/matplotlib/pyplot.py
Display all open figures.

Parameters
----------
block : bool, optional
```

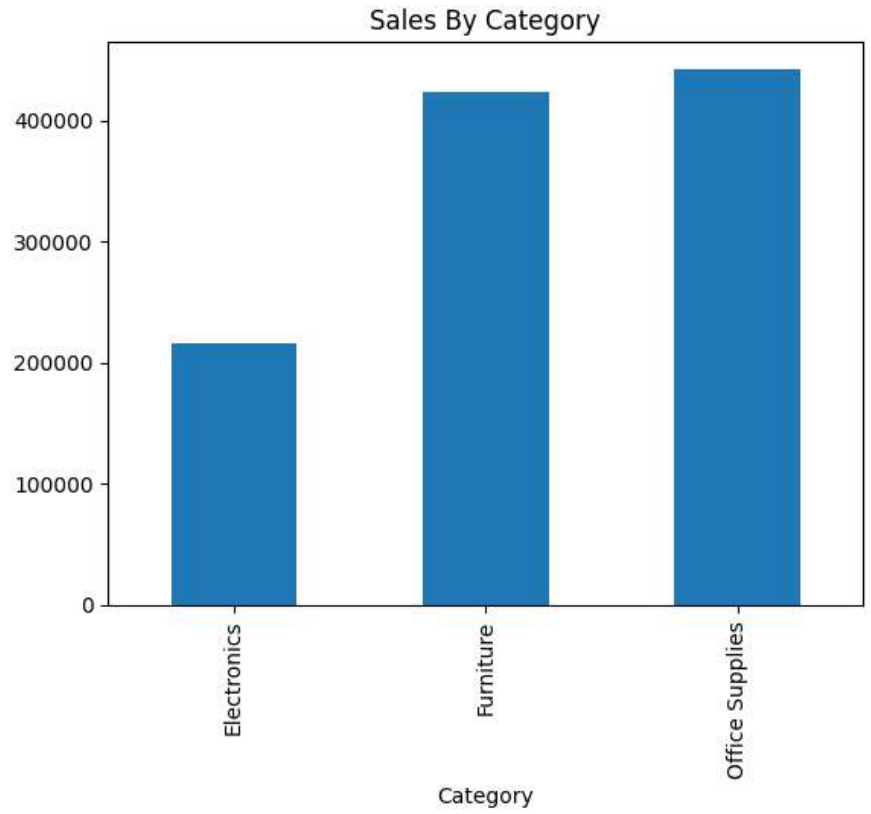


Answer: West rgion generated the highest sales

Q.2 Which product category generated the highest sales?

```
category_sales = df.groupby("Category")["Sales"].sum()
category_sales.plot(kind="bar")
plt.title("Sales By Category")
```

```
Text(0.5, 1.0, 'Sales By Category')
```

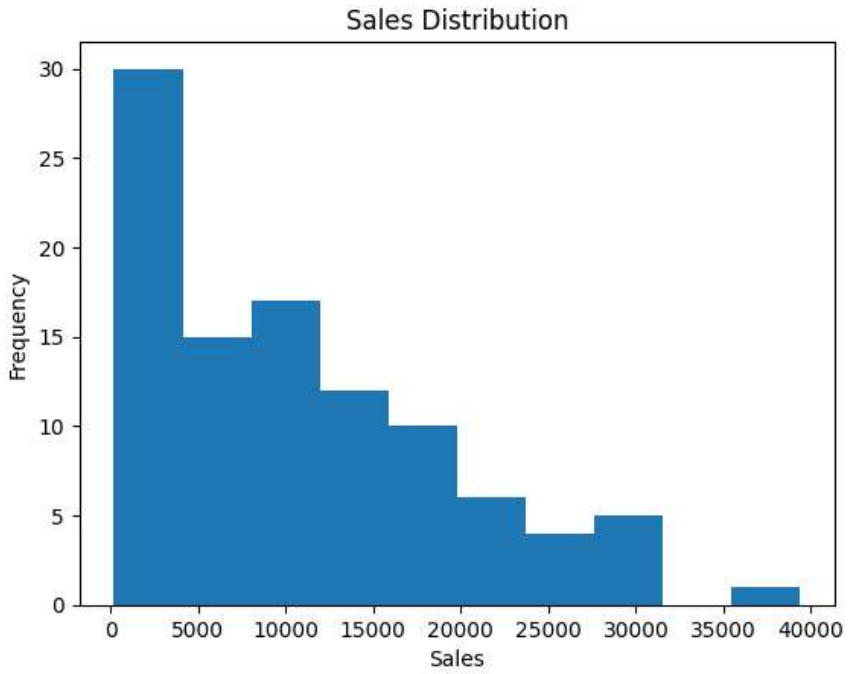


Answer: Office Supplies category generated the highest revenue

How are sales distributed?

```
plt.hist(df["Sales"])
plt.title("Sales Distribution")
plt.xlabel("Sales")
plt.ylabel("Frequency")
```

```
Text(0, 0.5, 'Frequency')
```

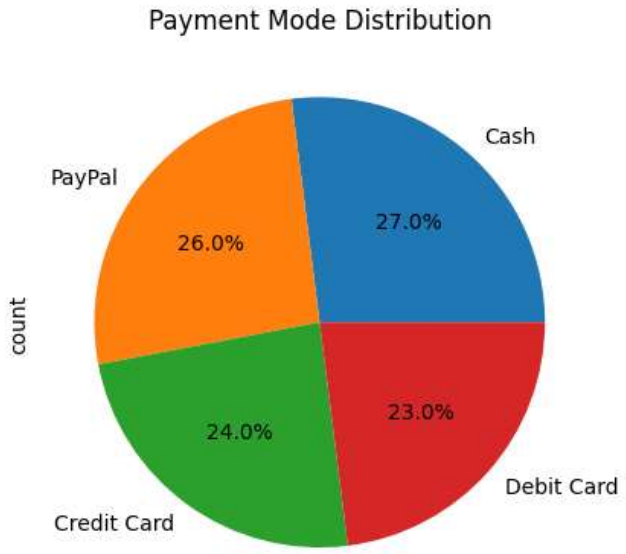


Answer: The sales is not evenlly distributed. Its inconsistent

Q.4 How are payment methods distributed?

```
payment = df["Payment_Mode"].value_counts()
payment.plot(kind="pie",autopct="%1.1f%%")
plt.title("Payment Mode Distribution")
```

```
Text(0.5, 1.0, 'Payment Mode Distribution')
```



Answer: The payment mode is not evenly distributed but there is no major inconsistency also.

Q.5 Which Cities generated the highest sales?

```
city_sales = df.groupby("City")["Sales"].sum()
city_sales.plot(kind="bar")
plt.title("Sales By City")
plt.xlabel("City")
plt.ylabel("Sales")
```

```
Text(0, 0.5, 'Sales')
```

